

Effect of artificial night lighting on temporally partitioned spiny mice

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We studied the effect of ecological light pollution on a rocky desert community, focusing on 2 spiny mouse congeners, nocturnal *Acomys cahirinus* (common spiny mouse) and diurnal *Acomys russatus* (golden spiny mouse). We hypothesized that in response to artificial illumination *A. cahirinus* will decrease its activity and *A. russatus* will increase its activity, and thus temporal overlap and interspecific competition could increase. Our study took place in 4 field enclosures: the 1st and 3rd months were controls with natural light, and in the 2nd month artificial illumination, simulating low levels of light pollution, was set for the first 3 h of the night. We implanted temperature-sensitive radiotransmitters to monitor mouse activity, and individual identification tags with automonitored foraging patches were used to track foraging behavior. *A. cahirinus* decreased activity and foraging with artificial lighting, restricting movement particularly in less-sheltered microhabitats, probably because of increased predation risk. Because illumination restricted both activity time and space, intraspecific encounters of *A. cahirinus* over foraging patches increased during and following the illuminated hours. However, diurnal *A. russatus* did not expand its activity into the illuminated hours, possibly due to the presence of competing *A. cahirinus*, or to nonfavorable environmental conditions. Therefore, overt interspecific competition was not affected by experimental light pollution. Light pollution had a negative influence by reducing overall activity and producing a relatively underexploited temporal niche, which may promote invasion of alien species that are less light sensitive; and by increasing intraspecific overlap in foraging *A. cahirinus*.

Key words: *Acomys cahirinus*, *Acomys russatus*, activity patterns, light night niche, light pollution

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Artificial illumination allows humans to expand their activity during the night. As people alter natural areas and build new settlements and infrastructures, artificial lighting increases. The phenomenon of artificial lights that alter natural light patterns in ecosystems was termed ecological light pollution by Longcore and Rich (2004). Any artificial light (e.g., streetlights, vehicles, and fishing boats), however low or sporadic, can be a source of such pollution. Conservative estimates suggest that in 2001 at least 18.7% of the terrestrial area of Earth was exposed to ecological light pollution (Cinzano et al. 2001), and it continues to rise with human populations and technological development.

Ecological light pollution is an increasing source of concern for ecologists. A sudden glare of artificial light can cause temporary blindness and damage night vision for a considerable period of time in mammals (Beier 2006; Lythgoe 1979) and amphibians (Buchanan 1993). Light sources can cause severe disorientation; well-publicized examples are lights that attract sea turtle hatchlings inland (Salmon 2003; Witherington and Martin 1996) and lights that cause millions of nocturnal migrating birds annually to crash into illuminated

buildings (Gauthreaux and Belser 2006; Ogden 1996). Artificial light also affects endogenous biological clocks (DeCoursey 1986) and activity patterns (Abramsky et al. 2004; Henderson and Powell 2001), thus affecting individual fitness (Dauchy et al. 1997). By altering species' activity patterns, ecological light pollution can alter interspecific interactions such as predation and competition (Frank 2006; Rydell 1992), and thus affect community structure (Longcore and Rich 2004).

Most knowledge of the effects of artificial light in nature derives from anecdotal observations and hypotheses. Very few studies have investigated the subject in experimental settings, and even fewer have attempted to test and experiment at the community level. Ecological light pollution, despite being widespread and potentially harmful to naturally occurring communities, has barely been studied (Longcore and Rich 2004).



We conducted an experimental study of the effect of artificial light pollution on a rocky desert community. Our research focused on 2 omnivorous congeners, the nocturnal common spiny mouse (*Acomys cahirinus*) and the diurnally active golden spiny mouse (*Acomys russatus*—Shargal et al. 2000). Such extreme temporal segregation between congeners is unusual (Roll et al. 2006) and appears to be a response of *A. russatus* to the presence of *A. cahirinus* (Gutman and Dayan 2005; Shkolnik 1971). We have been studying this system for almost 2 decades to gain insight into the physiological mechanisms controlling activity patterns (Elvert et al. 1999; Kronfeld-Schor et al. 2001; Levy et al. 2007), the roles of biotic interactions at 3 trophic levels, and abiotic pressures in the evolution and maintenance of temporal partitioning along the diel axis (Gutman and Dayan 2005; Jones and Dayan 2000; Jones et al. 2001; Kronfeld-Schor and Dayan 2003; Mandelik et al. 2003).

Clearly the cue of natural light plays a key role in the structure of this community. Research results in laboratory settings revealed that light regimes affect activity patterns and circadian clocks in both spiny mouse species (Cohen and Kronfeld-Schor 2006; Cohen et al. 2009; Haim et al. 1997; Weber and Hohn 2005). Moreover, previous research shows that even relatively low illumination during moonlit nights in the field (Mandelik et al. 2003), or its simulation in laboratory settings (Cohen et al. 2010; Eilam 2004; Mandelik 1999), affects microhabitat use and behavior of the spiny mouse during foraging.

We studied spiny mouse populations of both species under seminatural conditions in large field enclosures. We illuminated these enclosures artificially during the first 3 h of the night for 1 month and studied spiny mouse activity levels and foraging behavior, comparing them with the patterns that preceded and followed the illumination period. Our working hypotheses were as follows. *A. cahirinus* will reduce and restrict its activity in open microhabitats in response to light pollution (Mandelik et al. 2003). *A. russatus* will extend its activity into the night in response to light pollution and reduction of activity of its nocturnal congener. Previous studies show that when kept alone, individuals of *A. russatus* also become nocturnally active (Gutman and Dayan 2005; Shkolnik 1971). Finally, because the light cycle plays a major role in this system (Kronfeld-Schor and Dayan 2003), light pollution will disrupt this natural cycle and induce temporal overlap and interspecific competition. This experiment was designed to be the 1st to explore possible effects of artificial illumination at the day–night interface and at the community level.

MATERIALS AND METHODS

Study enclosures and test subjects.—Research took place at four 20 × 50-m enclosures located on the eastern slopes of the Judean Desert near the Ein Gedi nature reserve (31°28'N, 35°23'E, 300 m below sea level). Enclosures were constructed of 10-mm wire mesh buried 3 cm into the ground and standing

70 cm high. The upper 40 cm of both sides of the fence was covered with aluminum flashing to prevent mice from climbing over. The wire-mesh fence was permeable to spiny mouse predators (raptors, foxes, and snakes—Jones and Dayan 2000; Jones et al. 2001) and prey (vertebrates, seeds, and vegetation—Kronfeld-Schor and Dayan 1999); thus the enclosure structure provided seminatural conditions for our experiment. A rocky terrace ran along the middle of each enclosure providing continuous sheltered habitat, with an open habitat that offered no refuge on both sides. We captured 32 mature spiny mice, 16 individuals of each species, in the area around the enclosures, using Sherman live traps (LFA folding trap, 7.6 × 8.9 × 22.9 cm; H. B. Sherman Traps, Inc., Tallahassee, Florida). Spiny mice were marked individually with passive integrated transponder tags (details below), and each enclosure was populated with 8 individuals, 4 of each species, with a sex ratio of 1:1. A constant amount of food was added once a week (200 g of mouse pellets and mixed seeds), and water was available ad libitum (Gutman and Dayan 2005).

Experimental illumination.—The study lasted 3 consecutive months, and in the 2nd month the study area was illuminated for the first 3 h of the night (~1715–2015 h), beginning at twilight when natural light decreased below 0.3 lux, measured on the ground with a TES-1337 photometer (TES Electrical Electronics Corp., Taipei, Taiwan) to the nearest 0.01 lux. Illumination levels were reasonably constant throughout the enclosures at an average intensity of approximately 2 lux. This low light intensity was similar to light pollution measured about 40 m from an illuminated road junction in the study area. Enclosures were illuminated by 70-watt yellow metal halide lamps (Osram Ltd., Munich, Germany) on top of six 3-m-high poles. These lamps are similar in light color to frequently used high-pressure sodium-vapor lamps, generating realistic ecological light pollution in terms of light intensities and spectrum. Illumination was activated by a “super quiet” generator (E20; Honda Ltd., Rishon LeTsiyon, Israel) that was set at 150 m from the enclosures inside an insulated acoustic box, which rendered it noiseless at this distance.

Spiny mouse body temperatures.—Spiny mouse body temperature was monitored as a surrogate for mouse activity (Cohen and Kronfeld-Schor 2006; Decoursey et al. 1998; Levy et al. 2007). Body temperature radiotransmitters (Epx76 single-stage transmitters with 0.1°C precision, weight 3.8 g; Sirtrack Ltd., Hastings, New Zealand) were implanted in the abdominal cavity of 28 individuals (Levy et al. 2007). Each implanted transmitter uses a unique frequency. Signals were logged once every 18 min by 2 scanner-receivers (RX-900; Televilt Ltd., Stockholm, Sweden) connected to 3 dipole antennas. The receiver scanned each frequency for a period of 40 s, and whenever a signal was received, the time, frequency, pulse parameters, active antenna, and signal strength were logged. Data enabled tracking individual body temperature rhythms and activity (see Levy et al. [2007] for more technical details). Receivers were connected to a battery (450A; Schnapp Ltd., Netanya, Israel) charged during the day by a solar panel (SQ80; Shell Ltd., The Hague, Netherlands) and

worked constantly during the experimental period, recording about 3 million signals. A dedicated computer program was developed to filter noise and process this amount of data, and subsequently, processing was conducted through Access SQL queries (Microsoft Corp., Redmond, Washington).

Spiny mouse foraging behavior.—Spiny mice were marked by injecting passive integrated transponder tags (ISO FDX-B; AllFlex Ltd., Queensland, Australia) under their ventral skin, enabling individual identification when they mounted the automonitored foraging (AMF) patches set in the enclosures. AMF patches comprised a plastic plate (25 cm in diameter) in which 2 liters of local soil was mixed with 3 g of broken sunflower seeds and a transceiver (2001F-ISO; Biomark Ltd., Boise, Idaho) with a small round antenna (20 cm in diameter). The antenna was placed beneath the plate and connected with a cable to the transceiver, which logged the forager tag and the time (resolution in seconds), providing exact information about each mouse on the patch and the time and duration of its foraging. The transceivers in each enclosure were placed together in a brick box, and their antenna cables, leading to the plates in different microhabitats, were inserted into plastic tubes to prevent damage from animals and weather. Power supply was the same as for the above-mentioned body temperature receivers.

Three AMF patches were placed in each enclosure, each at a different microhabitat—under-boulder (UB) microhabitat (on the rocky terrace with overhead shelter), between-boulders (BB) microhabitat (on the terrace surrounded by large stones but no direct overhead cover), and open (O) microhabitat (at a distance of 1 m from the rocky terrace—Gutman and Dayan 2005). These microhabitats constitute a gradient in the degree of shelter, with UB the safest from avian predators and O the least protective.

During each experimental month a 4-day foraging session occurred, yielding in total ~100,000 logged foraging records. During the foraging sessions AMF patches were renewed to initial seed density 3 times a day, at 1st light, last light, and 3 h after last natural light. Data from the 1st day of each session were omitted from analyses because this period was used to acquaint mice with the AMF patches. Due to a shortage of 2 transceivers (2 devices malfunctioned), 1 enclosure contained only 1 AMF patch, which was positioned in the BB microhabitat. The other microhabitats in that enclosure contained simple artificial foraging patches but with no automonitoring equipment.

In addition, AMF patches were activated without seeds for 7–8 days each month to detect mouse foraging and presence in the patches when no food was available. This monitoring was conducted only at the 3 fully equipped enclosures. A total of ~6,000 readings were recorded in the AMF patches without seeds during the experiment.

Study design.—Spiny mouse trapping and introduction into the enclosures was completed approximately 1 month prior to the beginning of the experiment, allowing mice time to acclimate. The experiment was conducted during late autumn and winter (mid-October to mid-January, 2007–2008). The 1st

and last months of the experiment served as controls, with natural light regime, and in the 2nd month illumination was set for the first 3 h of the night. Body temperatures were recorded during the entire period. During the last week of every month the foraging experiment was conducted during the new moon phase to maximize the difference between experimental lighting conditions. We also activated the AMF patches without seeds for 8 days during the 1st and 2nd months and for 7 days during the last month of the experiment. All experiments were conducted with the approval of the Israel Nature and Parks Authority (permit 2007/28812) and animal handling and care were consistent with guidelines of the American Society of Mammalogists (Gannon et al. 2007).

Activity and body temperature thresholds.—To determine body temperature threshold for activity we used concurrent records (within <5 min) of an individual's body temperature and foraging visits in the AMF patches. These records enabled us during each experimental month to calculate average body temperature of each individual while it foraged. This average minus 1 *SD* was defined as the activity body temperature threshold of that individual, above which it was considered active. The average threshold (mean $\pm$ *SE*) was 37.1°C $\pm$ 0.2°C for *A. cahirinus* and 36.3°C $\pm$ 0.2°C for *A. russatus*.

Data analysis.—We compared mouse activity and foraging behavior during the first 3 h of the night (light manipulation hours) between experimental months. Mice were considered active when their body temperatures were above their monthly individual body temperature activity threshold. Activity also was estimated using signal strength differences, where a difference of >2 units of signal strength between consecutive records for an individual was taken to indicate different locations and thus to reflect activity (Elvert et al. 1999; Levy et al. 2007).

We analyzed only activity levels of individuals that were recorded during the entire experimental period (8 *A. cahirinus* and 13 *A. russatus* of 13 and 15 implanted individuals, respectively). Data from days in which food was supplied (for feeding or foraging experiments) and the 2 days following them were omitted from analysis because we detected a significant effect of food on mouse body temperature rhythms. Days in which researchers interfered also were removed. Activity levels in the hours of light manipulation were examined in a relative rather than absolute way to neutralize ambient temperatures, which constantly decreased during the experiment, and to account for possible changes in the quality of transmitter reception. Percentage of activity during the light manipulation hours was calculated for each mouse by dividing activity records in these hours by activity records during the entire day. To test for illumination effects we compared mouse relative activity levels between the 3 months of the experiment using 1-way analysis of variance (ANOVA) with repeated measures, with individuals as the repeated factor (Tukey honestly significant difference post hoc test, $\alpha = 0.05$).

Two metrics were derived from the AMF patches data: foraging duration (the accumulated number of logged records) and number of visits. A visit was defined as each time an

individual had entered a patch until it left the patch. A record was considered a new visit only if the individual was not present on the patch for at least 1 min earlier. Foraging also was examined in a relative manner; the percentages of foraging duration and foraging visits during the light manipulation hours were calculated as foraging duration and number of foraging visits during the light manipulation hours divided by foraging duration and foraging visits during the whole day, respectively. We examined foraging with seeds replenished 3 times a day; mice that were recorded only in 1 of the 3 experimental months were omitted from the analysis. Because of an extra AMF patch in BB microhabitat, and because 3 individuals in the O microhabitat were omitted from analysis, numbers of individuals examined differ between microhabitats. Thus, each microhabitat was analyzed separately. For each microhabitat we compared the foraging parameters between the 3 months of the experiment using 1-way ANOVA with repeated measures, with individuals as the repeated factor (Tukey honestly significant difference post hoc test, $\alpha = 0.05$). We applied sequential Bonferroni correction (Holm 1979) on the different microhabitat tests regarding the same foraging parameter to deal with multitests alpha degradation (Longland and Price 1991).

We also monitored mouse presence in the AMF patches when they contained no food. In the absence of food mice did not stay long in the patches, and thus the 2 foraging parameters, duration and number of visits, were equivalent. Therefore, we presented only the analysis of the results for the number of visits. We examined only individuals that were recorded in at least 2 of the 3 experiment months. O microhabitat was omitted from analysis because of too few data. We used a repeated-measures 2-way ANOVA, with illumination and microhabitat as main factors, and individuals as repeated factors (microhabitats were analyzed together because sample size did not differ between them).

We compared foraging activity overlap and interindividual encounters on the AMF patches between experimental months. An encounter was defined as each time 2 individuals were recorded in the same patch within a time difference of ≤ 2 s. For each mouse the percentage of encounters was calculated as the number of encounters divided by the total number of foraging logs, and compared between experimental months. To examine mouse encounters further during illumination we tested whether they were more concentrated in the sheltered (UB) microhabitat. For that we calculated the percentage of UB encounters from all encounters (at all microhabitats) during the light manipulation hours and compared this parameter between experimental months. In the same fashion we compared the percentage of UB foraging visits. In addition, level of aggression during encounters was examined. An encounter was considered aggressive if it resulted in immediate withdrawal of 1 of the foragers. A nonaggressive encounter occurred when the 2 mice stayed on the patch and were recorded together once again within < 10 s. Percentages of nonaggressive encounters from total encounters in the hours of the light manipulation were compared

TABLE 1.—Effect of illumination on activity and foraging of *Acomys cahirinus* during the hours of light manipulation. A summary of 1-way repeated-measures ANOVAs with illumination as the main factor and individuals as repeated factors, conducted separately for each row. Last 2 rows display the results of a 2-way repeated-measures ANOVA with microhabitat as the additional factor. Microhabitat abbreviations are: UB, under boulders; BB, between boulders; O, open habitat. NS denotes nonsignificance ($P > 0.05$) after applying sequential Bonferroni correction.

Parameter	d.f.	F	P
Relative activity level (body temperature)	2, 14	11.91	< 0.001
Relative activity level (signal strength)	2, 14	4.70	0.027
Relative foraging duration			
All microhabitats	2, 26	1.54	0.233
UB microhabitat	2, 18	1.28	0.302
BB microhabitat	2, 26	3.70	0.039NS
O microhabitat	2, 10	3.23	0.083
Relative foraging visits			
All microhabitats	2, 26	7.61	0.003
UB microhabitat	2, 18	4.40	0.028
BB microhabitat	2, 26	5.82	0.008
O microhabitat	2, 10	5.79	0.021
Relative foraging visits without food			
Light	2, 20	13.73	< 0.001
Light $\times$ Microhabitat	2, 20	2.99	0.073

between experimental months. All comparisons mentioned in this paragraph were conducted using 1-way ANOVA with repeated measures, with individuals as the repeated factor (Tukey honestly significant difference post hoc test, $\alpha = 0.05$).

In all of our repeated ANOVAs data were tested for normal distribution using the Kolmogorov–Smirnov test. The entire statistical analysis was conducted with STATISTICA software (version 7.1; StatSoft Inc., Tulsa, Oklahoma).

RESULTS

Analysis of activity levels, estimated from body temperature levels, revealed that illumination had a significant effect on *A. cahirinus*. Percentage of activity during the 3 h of light manipulation from all-day activity records was significantly lower during the illumination phase in comparison to both control months (Table 1; Fig. 1). Analyzing activity from signal strength differences, we revealed a similar pattern, but the difference in percentage of activity for *A. cahirinus* during the 3 h of light manipulation from all day activity records was significant only in comparison to the 1st control month (mean $\pm$ SE for illumination month: $13.0\% \pm 0.7\%$; for control 1: $17.8\% \pm 2.1\%$; for control 2: $15.9\% \pm 1.1\%$).

We examined foraging behavior of *A. cahirinus* with food supplemented. We found no significant effect of illumination on relative foraging duration of *A. cahirinus* during the manipulated hours (duration of foraging during the 3 h of light manipulation versus duration of foraging during the whole

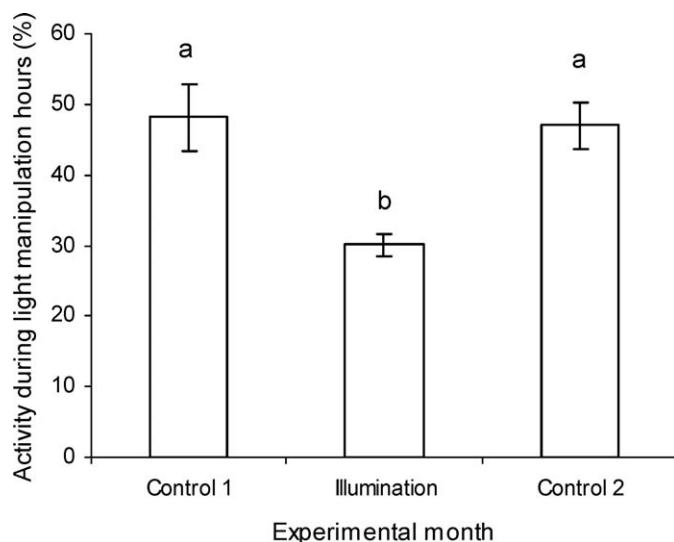


FIG. 1.—Activity of *Acomys cahirinus* during the hours of light manipulation versus all day (%) in each month (mean $\pm$ SE). Lowercase letters denote significant differences ($n = 8$, Tukey honestly significant difference, $P < 0.01$). Activity level was calculated as the number of body temperature records above the individual calculated activity threshold.

day) at all microhabitats (Table 1), although there was a trend of lowest foraging duration under illumination at all microhabitats except UB. However, illumination had a significant effect on the relative number of foraging visits (number of visits during the 3 h of light manipulation versus all day visits) in the patches in all microhabitats (Table 1). In all microhabitats together and in the BB microhabitat alone percentages of visits during the illumination phase by *A. cahirinus* were significantly lower than in both control months, in the O microhabitat percentages of visits during the illumination phase were significantly and marginally significantly lower compared to control 1 and control 2 months, respectively, and in the UB microhabitat percentages of visits during the illumination phase were significantly lower than 1 of the control months (Fig. 2). An interaction between illumination and microhabitat was evident; in the less-sheltered microhabitats the decrease in percentage of visits in the illumination month was greater.

In the absence of food in the foraging patches, results were similar to the above-mentioned visit results with food supplemented. Illumination had a significant effect on foraging visits of *A. cahirinus* in the hours of light manipulation; percentages of visits during the manipulated hours were significantly lower in the illumination month compared to both control months (Table 1). We also found a marginally nonsignificant interaction between illumination and microhabitat (Table 1); at both microhabitats analyzed (UB and BB) percentages of visits decreased during the illumination month, but this decrease was stronger and significantly different from both control months only at the BB microhabitat (mean $\pm$ SE for illumination month: $21.1\% \pm 4.9\%$; for control 1: $60.9\% \pm 6.1\%$; for control 2: $61.1\% \pm 8.5\%$). At the UB microhabitat illumination month was

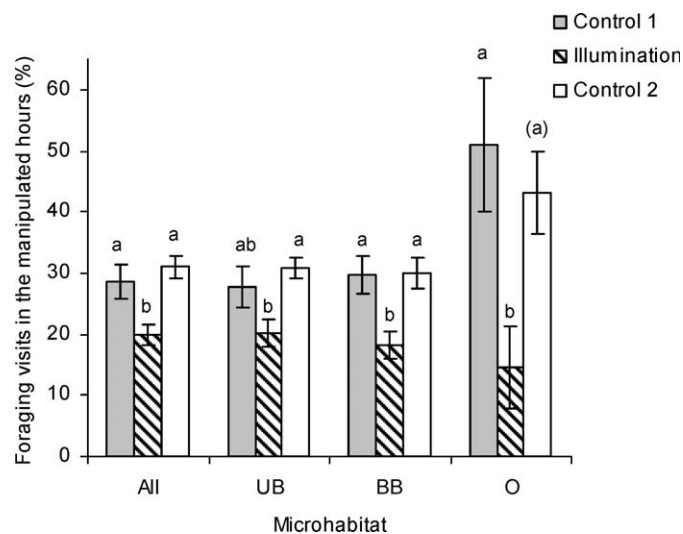


FIG. 2.—Foraging visits by *Acomys cahirinus* during the hours of light manipulation versus all day (%) in each month at the different microhabitats (mean $\pm$ SE): UB, under boulders; BB, between boulders; O, open habitat. Different letters above the columns denote significant differences within a microhabitat (Tukey honestly significant difference, $P < 0.05$), whereas brackets around them indicate marginal nonsignificance ($P < 0.07$). $n = 14, 10, 14$, and 6 for microhabitats All, UB, BB, and O, respectively. Note that foraging parameters were described by percentage rather than absolute values, hence the deceptively high figures for the O microhabitat where foraging was lowest.

significantly different only in comparison to the 1st control month (illumination: $24.4\% \pm 5.0\%$; control 1: $61.0\% \pm 7.5\%$; control 2: $38.3\% \pm 8.6\%$). During the days when seeds were supplemented in the foraging patches percentages of foraging visits of *A. cahirinus* during the hours of light manipulation (regardless of illumination) were lower than with days when no seeds were supplemented.

During both control months 1 clear peak in body temperature occurred during the hours of light manipulation (Figs. 3a and 3c), whereas during the illumination month this peak was reduced and an additional peak emerged after illumination ceased (Fig. 3b). To examine this new activity peak we tested foraging behavior during the hour after illumination ceased and found a significant effect of illumination on foraging duration ($F_{2,26} = 7.35$, $P = 0.003$). Percentage of foraging during the hour after illumination was significantly higher in comparison to the same hours in both control months ($P < 0.05$; mean $\pm$ SE for illumination month: $21.34\% \pm 2.08\%$; for control 1: $13.55\% \pm 2.26\%$; for control 2: $12.36\% \pm 1.91\%$).

This increase in foraging following illumination yielded an unexpected outcome: the percentage of encounters between foraging *A. cahirinus* was significantly higher during the illumination month in comparison to both control months (Table 2; Fig. 4). This pattern was particularly marked during the hour following illumination but also was found during the 2 other time intervals examined, all day and the hours of light manipulation.

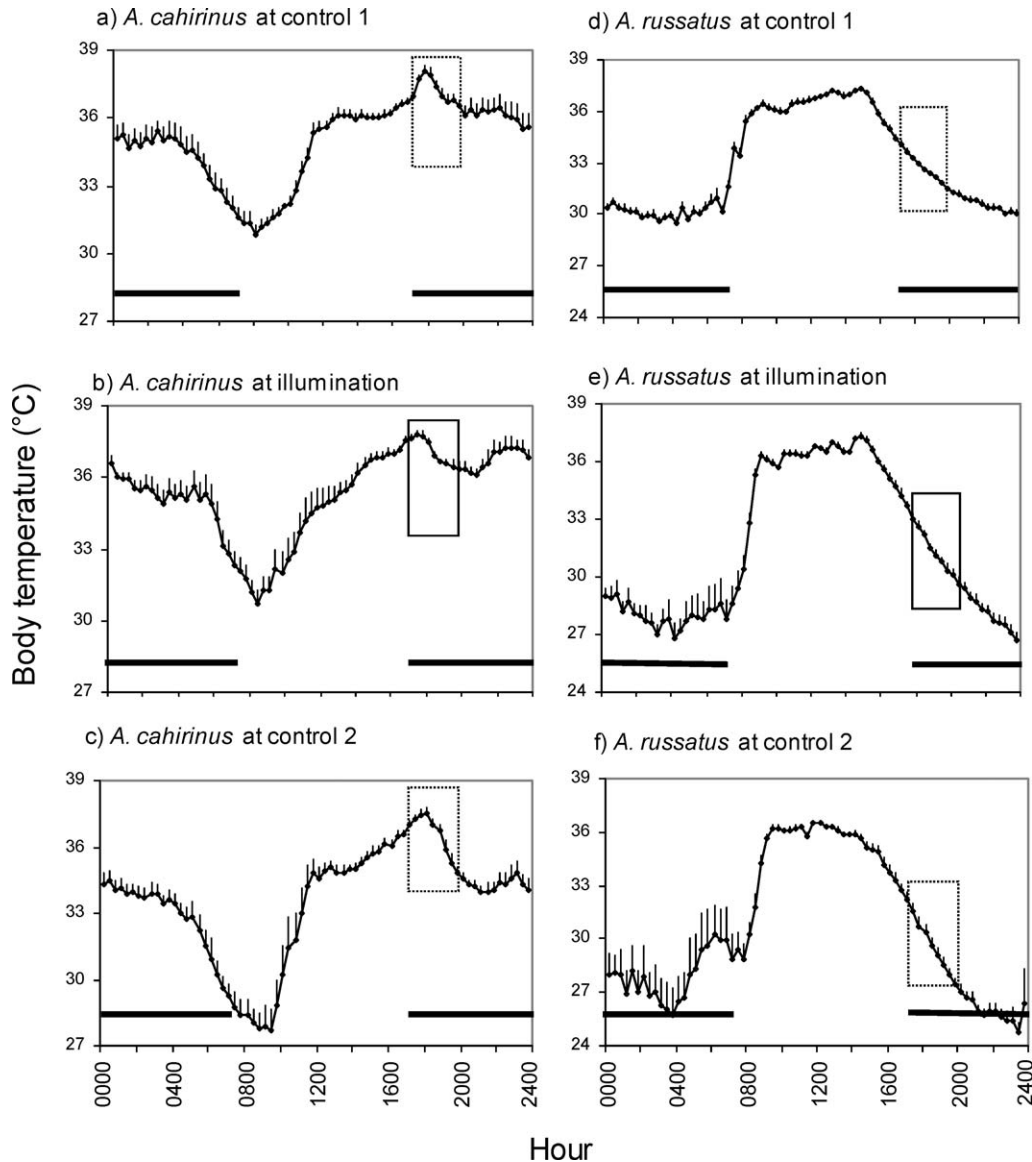


FIG. 3.—Mean daily body temperatures ($\pm SE$) during each experimental month for each species (8 *Acomys cahirinus* and 13 *A. russatus*). Data points were derived from individuals' means, which were calculated from the last 3 days of each experimental month. Rectangles (dashed for control and solid for illumination) mark the hours of light manipulation. Horizontal line indicates the darkness hours. a–c) *A. cahirinus* at control 1, illumination, and control 2 months, respectively. d–f) *A. russatus* at each experimental month. Note that the constant decrease in resting body temperatures within species along experimental months is attributed to the correspondingly constant decrease in ambient temperatures.

Illumination had a significant effect on the microhabitat distribution of the encounters ($F_{2,14} = 3.76$, $P = 0.049$). Under illumination encounters were more concentrated at the UB microhabitat (illumination month: $85.84\% \pm 7.70\%$; control 1: $71.19\% \pm 13.15\%$; control 2: $60.41\% \pm 8.84\%$), where the difference between illumination and control 2 months was significant ($P = 0.04$). However, foraging visits showed only a trend for higher percentage of UB visits under illumination ($F_{2,18} = 1.60$, $P = 0.23$; illumination month: $51.82\% \pm 7.63\%$; control 1: $42.34\% \pm 6.89\%$; control 2: $44.25\% \pm 3.21\%$). No effect of illumination was found on the percentages of nonaggressive encounters versus

total encounters in the hours of the light manipulation ($F_{2,18} = 0.64$, $P = 0.54$).

Individual *A. russatus* were not active (i.e., no body temperature records were above their individual monthly activity threshold) during the hours of light manipulation (first 3 h of the night) in all experimental months. Moreover, *A. russatus* rarely (24 of 2,688 recorded visits, 23 of them during the 1st month) were recorded in the foraging patches during the light manipulation hours. Thus, illumination had no effect on activity and foraging behavior of *A. russatus*.

The pattern of mean body temperature rhythms of 13 *A. russatus* during the last 3 days of each month was similar for

TABLE 2.—Effect of illumination on intraspecific encounters of *Acomys cahirinus* over 3 different time intervals. A summary of 1-way repeated-measures ANOVAs with illumination as main factor and individuals as repeated factor, conducted separately for the different time intervals.

Encounters time intervals	<i>d.f.</i>	<i>F</i>	<i>P</i>
All day	2, 26	13.50	< 0.001
Manipulated hours	2, 26	6.99	0.004
Hour following the manipulated hours	2, 26	15.43	< 0.001

all individuals. Body temperatures decreased constantly during the hours of light manipulation throughout all months, and peaks in body temperatures occurred exclusively during the day (Figs. 3d–f).

We found no overlap in interspecific activity in the hours of light manipulation, and only a negligible 7 interspecific encounters on the AMF patches. Thus, we observed no effect of illumination on direct interactions between the 2 spiny mouse species.

DISCUSSION

We asked how artificial light pollution might affect the activity patterns and community structure of temporally partitioned species. As expected, artificial light pollution reduced activity and foraging behavior of nocturnal *A. cahirinus*. It is well known that illumination elevates the actual and perceived predation risk by nocturnal raptors (Brown et al. 1988; Clarke 1983; Lima and Dill 1990; Lockard and Owings 1974). A previous study showed that spiny mouse foraging was less profitable during moonlit nights (Mandelik et al. 2003), as has been described for other nocturnal rodents (Beier 2006). Similar results were demonstrated under artificial illumination in the field (Bird et al. 2004; Brown et al. 1988; Kotler 1984a). The reduction in foraging and activity under experimental lighting conditions in our study probably reflects the increase in perceived predation pressure.

Light pollution affected foraging of *A. cahirinus* primarily in the less-sheltered microhabitats where risk of predation is expected to be greatest, especially under conditions of elevated illumination (Brown et al. 1988; Kotler 1984b; Longland and Price 1991). Similar responses of nocturnal desert rodents to increased light intensities were described in many studies of foraging under moonlight (Bowers 1988; Lockard and Owings 1974; Mandelik et al. 2003) and under artificial illumination (Abramsky et al. 2004). These studies relied only on levels of foraging activity in passive artificial patches. A behavioral study carried out in controlled laboratory conditions revealed that under elevated illumination spiny mice reduced their overall activity to a much greater extent than their actual foraging (Mandelik 1999), so by also studying activity patterns, our study gauges actual changes in activity rhythms at a much finer scale.

The relative number of foraging visits was lower under artificial illumination, but relative foraging duration in the

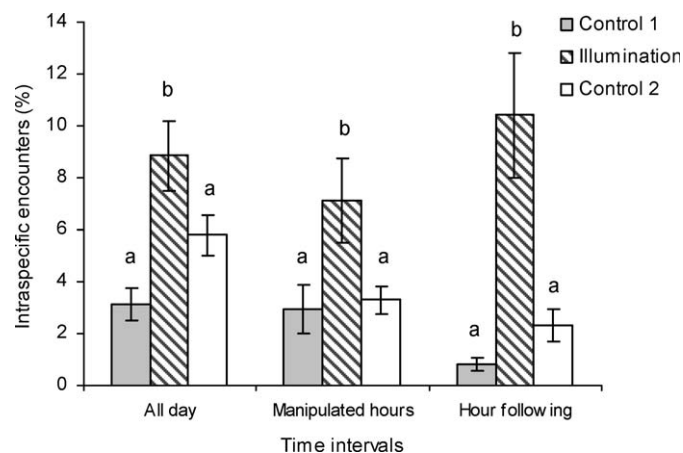


FIG. 4.—Intraspecific encounters among *Acomys cahirinus* as percent (mean $\pm$ SE) of foraging duration in each month over different time intervals: Manipulated hours constitute the first 3 h of the night; Hour following is the hour following the manipulated hours. Control 1 and 2 represent 1st and last of the 3 experimental months, respectively. Different letters above the columns denote significant ($P < 0.05$) differences within a time interval (Tukey honestly significant difference). $n = 14$ (for all groups).

artificial patches was reduced but not significantly so. Mandelik et al. (2003) found a similar pattern in moonlit nights and suggested that the movement to a foraging patch is riskier than actual foraging once in a patch, because owls usually attack only moving prey (Kaufman 1974; Metzgar 1967). While foraging in a patch the mouse is relatively motionless, and it also is partly camouflaged by the mosaic of shadows created by the rocky desert terrain (Mandelik et al. 2003).

We observed differences in magnitude in the response to illumination using the 2 different measures of activity (body temperatures and signal strength differences), probably reflecting differences in acuity of the 2 measures: signal strength differences are a less sensitive measure because they are based upon changes in location whereas an animal can be active without changing location. Similarly, we observed differences in the magnitude of the foraging response to illumination with and without seed supplementation, probably reflecting the influence of seed renewal times on foraging patterns. Nevertheless, the patterns emanating from illumination, of reduced activity and foraging, were consistent in all cases.

Acomys cahirinus is active mostly during the early night hours (Elvert et al. 1999; Kronfeld et al. 1994; Mendelssohn and Yom-Tov 1987). Body temperature rhythms revealed that our experimental light pollution changed the shape of activity rhythms, with a new peak after the hours of artificial lighting, replacing or adding to the normal peak in the early hours of the night. It appears that *A. cahirinus* compensated for reduced activity during illumination with increased activity and foraging following it. Such a behavioral response has an obvious adaptive value, and it is expected by the risk allocation hypothesis (Lima and Bednekoff 1999). Shifting activity time in response to high predation risk has been found

in some other prey species (Abramsky et al. 1996; Daly et al. 1992; Sih and McCarthy 2002).

The change in activity and foraging was coupled with an increase for *A. cahirinus* in intraspecific encounters on the foraging patches during the illumination month. Because most encounters result in immediate withdrawal of 1 of the foragers (>80%), suggesting intraspecific interference, we hypothesize that increased encounter rate reflects population-level elevation in interference competition. It appears that illumination reduced the amount of low-risk activity time, thus increasing competition among *A. cahirinus*, especially during the hour after illumination ceased when mice were eager to forage. Surprisingly, a similar pattern was found when the hours of illumination alone were compared between experimental months. Our results imply that illumination shifted habitat use to the more sheltered microhabitats, and that more concentrated foraging activity in 1 microhabitat could be the cause for the increase in intraspecific competition in *A. cahirinus* under illumination. Abramsky et al. (2004) found more habitat overlap between 2 competitive gerbil species under illumination. They suggested that under illumination individuals were more focused on the increased predation risk and thereby became less aggressive competitors. Decrease in intraspecific aggression might result in more foraging overlap (more encounters); however, this does not appear to be the case in our study.

Diurnal *A. russatus* did not expand their activity into the night in response to experimental light pollution, perhaps because activity of *A. cahirinus* was reduced but not decimated. Previous studies have shown that *A. russatus* is active during the night only in the absence of *A. cahirinus*, and even then, only part of the activity shifted to the night (Gutman and Dayan 2005; Shkolnik 1971). Moreover, illumination also produced less favorable activity conditions for *A. russatus* because of elevated predation risk. Also, our research was conducted in winter, and during the illumination phase average ambient surface temperature of 21.5°C was recorded. *A. russatus* regulates normal body temperature only between 22°C and 35°C (Shkolnik and Borut 1969). However, nocturnal activity was not recorded even when ambient temperature during the illumination phase was >22°C (maximum of 24°C). It is also possible that the low intensity of the experimental illumination was not enough to trigger activity for a diurnally active species. Nevertheless, *A. russatus* did not expand its activity into the illuminated night period, so no increase in interspecific overlap in foraging times, and therefore no overt indication of increased interspecific competition, was seen.

Overall, light pollution caused a reduction of spiny mouse activity. *A. cahirinus* reduced its activity, whereas *A. russatus* did not take advantage of the illuminated hours by increasing its activity. Previous studies referred to artificial light as a new nocturnal niche, the light night niche (Garber 1978; Henderson and Powell 2001). Our experimental light pollution created such a relatively underexploited temporal niche, less attractive to one of the local species and entirely unexploited by the

other. Such conditions could promote the introduction or spread of alien species that are less sensitive to illumination, such as the house mouse. A conceptually similar phenomenon was described in Pacific geckos, where findings suggest that artificial illumination had a major role in the exclusion of some native gecko species by the invasive Asian house gecko (Newbery and Jones 2007; Petren et al. 1993). Arlettaz et al. (2000) proposed that artificial light might mediate the exclusion of the lesser horseshoe bat in western Europe by its competitor (common pipistrelle).

In summary, light pollution reduced activity of *A. cahirinus*, restricted its foraging, and elevated intraspecific interference competition. Diurnal *A. russatus* did not expand its activity into the illuminated phase, thus no increase in interspecific overlap resulted, leaving a fairly underexploited time window in this 2-species system. These results add to a growing volume of scientific evidence of adverse effects of ecological light pollution on wildlife.

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